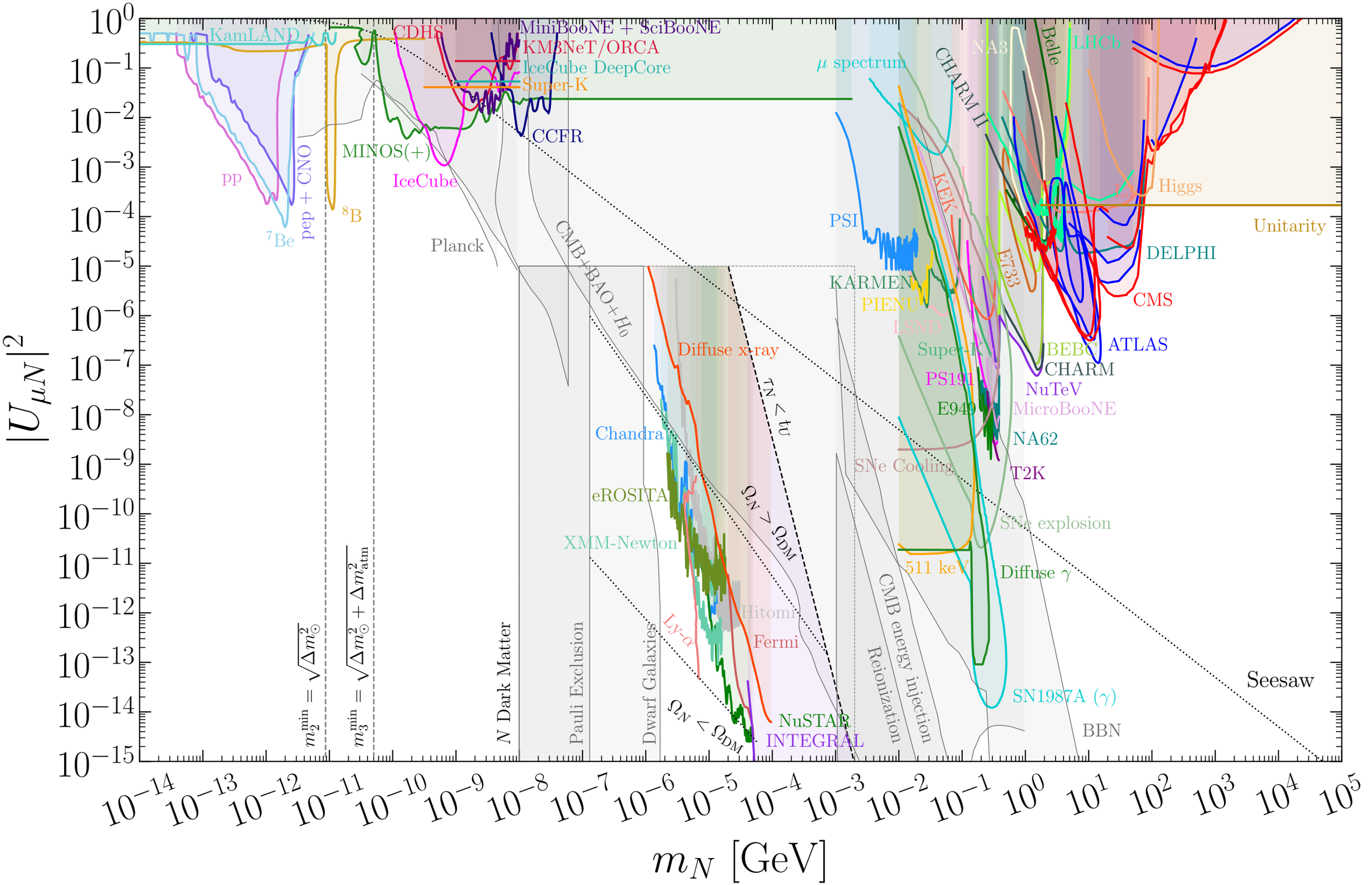

$|U_{\mu N}|^2$
m_N [GeV]
KamLAND
pp
^{7}Be
pep + CNO
^{8}B
CDHS
MINOS(+)
IceCube
MiniBooNE + SciBooNE
KM3NeT/ORCA
IceCube DeepCore
Super-K
CCFR
Planck
CMB+BAO+H_0
μ spectrum
CHARM II
NA3
Belle
LHCb
PSI
KEK
E733
Higgs
Unitarity
DELPHI
CMS
ATLAS
KARMEN
PIENU
LSND
Super-K
BEBC
CHARM
NuTeV
PS191
E949
MicroBooNE
NA62
T2K
SNe Cooling
SNe explosion
Diffuse γ
511 keV
SN1987A (γ)
BBN
Seesaw
Diffuse x-ray
$\tau_N < t_U$
$\Omega_N > \Omega_{DM}$
Chandra
eROSITA
XMM-Newton
Ly-α
Hitomi
Fermi
NuSTAR
INTEGRAL
$\Omega_N < \Omega_{DM}$
N Dark Matter
Pauli Exclusion
Dwarf Galaxies
CMB energy injection
Reionization
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$